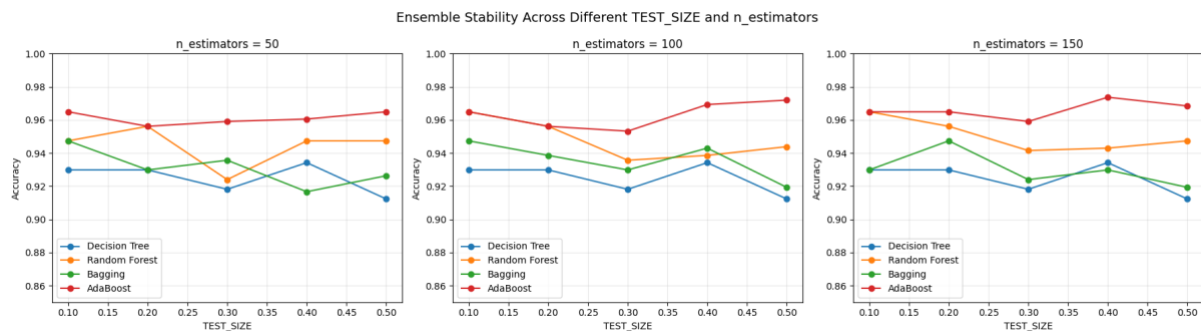


Assignment3_20201118_Analysis

1. Ensemble Learning

Q1. Change the test size hyperparameter and observe which ensemble method remains the most stable when the dataset is relatively small. Experiment by varying n estimators.

The TEST_SIZE value was varied from 0.1 to 0.5 and the experiment was repeated for $n_{\text{estimators}} = 50, 100, \text{ and } 150$. The goal was to observe how each ensemble model reacts as the training set becomes smaller. The accuracy trends are summarized in the figure below.



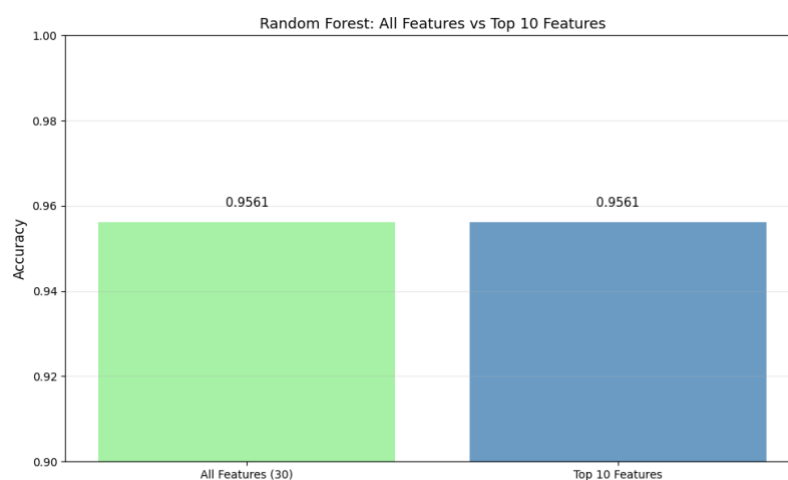
Overall patterns from the results are as follows.

- Decision Tree shows the largest fluctuation. Its accuracy decreases consistently when TEST_SIZE increases, indicating strong sensitivity to reduced training data.
- Bagging is more stable than Decision Tree but still displays noticeable variation, especially around mid-range TEST_SIZE splits.
- Random Forest maintains relatively consistent accuracy across all settings, with only small changes as TEST_SIZE grows.
- AdaBoost is the most stable model. It sustains high accuracy and shows minimal variation, even improving slightly at some TEST_SIZE values.

From these observations, AdaBoost remains the most stable ensemble method under reduced training data, followed by Random Forest. Decision Tree is the most unstable model in this setting.

Q2. Select the 10 most important features from the Random Forest, visualize them, and analyze how the accuracy changes when training with only these features.

The ten most important features were selected from the Random Forest model using its feature importance scores. These features include worst area, worst concave points, worst radius, mean concave points, worst perimeter, mean perimeter, mean radius, mean concavity, mean area, and worst concavity. After selecting these variables, the Random Forest classifier was retrained using only this reduced set, and its performance was compared with the model trained on all thirty features.



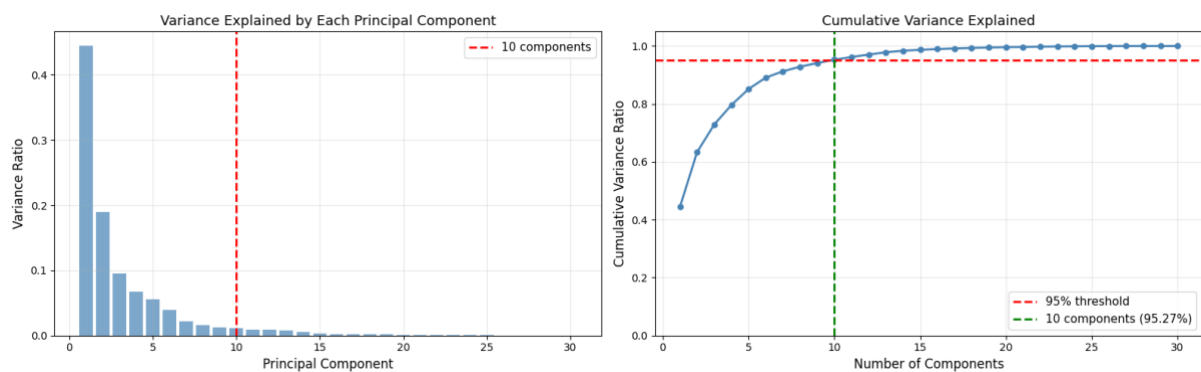
The accuracy of the full-feature model was 0.9561, and the model trained with the top ten features achieved the same accuracy of 0.9561. This result shows that the reduced feature set retains all essential information required for the Random Forest to perform effectively. Because the discarded features contribute very little to the decision-making process, the model maintains identical performance while using only one-third of the original dimensionality.

In summary, selecting the top ten features yields a more compact model without any loss in accuracy, indicating that these features capture the dominant patterns necessary for classification in this dataset.

2. Dimension Reduction

Q1. How many principal components are required to retain more than 95% of the variance?

Principal Component Analysis was applied to determine how many components are needed to retain more than 95 percent of the total variance in the dataset. The cumulative variance curve and the per-component variance distribution are shown below.



The first principal component accounts for 44.41 percent of the variance, and the second component adds an additional 18.94 percent. The cumulative variance continues to rise gradually with each added component. The threshold of 95 percent is exceeded when the first ten components are included, reaching a total of 95.27 percent.

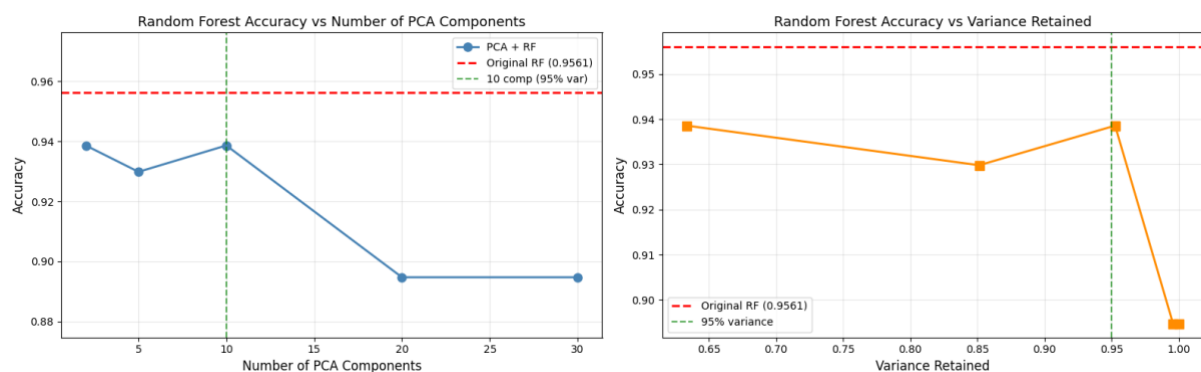
Key cumulative values are summarized as follows.

- PC1 to PC2: 63.36 percent
- PC1 to PC5: 85.14 percent
- PC1 to PC10: 95.27 percent

Therefore, ten principal components are required to retain more than 95 percent of the original variance, and this reduced representation preserves almost all essential structure in the dataset.

Q2. Apply Random Forest after PCA and compare the performance. Set n_estimators = 100.

Random Forest was trained on PCA-transformed data using $n_estimators = 100$, and its accuracy was compared with the original model trained on all 30 features. PCA component counts were selected to represent different levels of variance retention, ranging from 63 percent to 100 percent. The corresponding accuracy values are shown in the figure below.



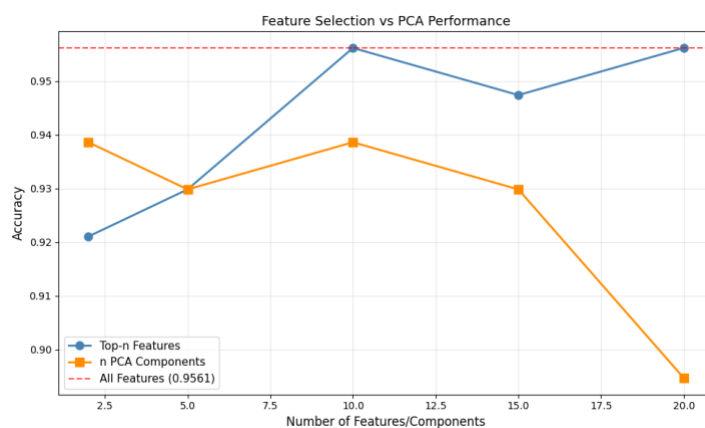
Using a small number of components preserves a substantial portion of the model's performance. With two components (63.36 percent variance), the accuracy is 0.9386, and with ten components (95.27 percent variance), the accuracy remains the same at 0.9386. Accuracy with five components is slightly lower at 0.9298.

However, when the number of PCA components is increased to twenty or thirty, the accuracy decreases noticeably to 0.8947 despite retaining nearly all variance. This drop occurs because PCA rotates the feature space, and the resulting components no longer align well with the axis-based splits used by tree models.

Overall, the Random Forest trained on PCA features performs reasonably well when using a small number of principal components, but it does not outperform the original model without PCA. The baseline accuracy of 0.9561 remains the highest among all tested configurations.

Q3. Compare the performance between using the top n features selected by Random Forest feature importance and using n PCA components.

Random Forest performance was compared under two dimensionality-reduction strategies: selecting the top n features based on feature importance, and using the first n principal components obtained from PCA. Accuracy values for $n = 2, 5, 10, 15$, and 20 are summarized in the figure below.



The top-n feature approach shows a clear advantage as n increases. With ten features, the accuracy reaches 0.9561, matching the performance of the full model. Even with fifteen or twenty features, the accuracy remains high, staying close to the original baseline. This indicates that the variables identified as most important by Random Forest preserve class-relevant information effectively.

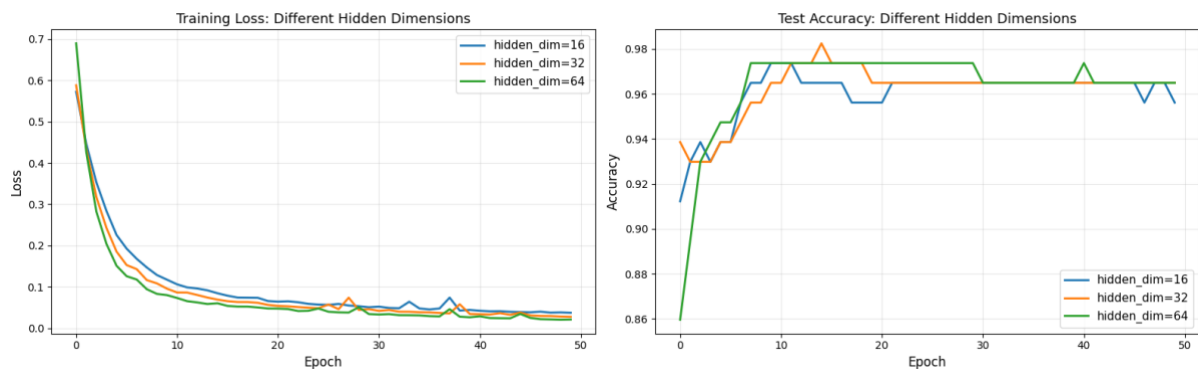
In contrast, using n PCA components yields lower accuracy for most values of n . The PCA-based model performs reasonably well for small n , but the accuracy does not improve as n increases. For $n = 20$, even though PCA retains almost all variance, the accuracy drops to 0.8947. This pattern occurs because PCA maximizes variance without considering class separation, and the resulting components may not align well with the decision boundaries used by tree-based models.

Overall, top-n feature selection provides more consistent and higher accuracy than PCA for the same dimensionality, demonstrating that selecting features directly relevant to the classification task is more effective than retaining variance alone.

3. Neural Network

Q1. Change hidden dim to 16, 32, and 64, and find which configuration achieves the best performance.

The MLP was trained with hidden dimensions of 16, 32, and 64 to compare how model capacity affects convergence and classification accuracy. The training loss curves and test accuracy trends are shown below.



All three configurations converge smoothly, but their performance differs slightly. The model with 16 hidden units achieves stable training but reaches a best accuracy of 0.9737.

Increasing the hidden dimension to 32 leads to the highest recorded accuracy of 0.9825, indicating that this setting provides enough capacity to capture the structure of the data without overfitting. The 64-unit model performs similarly and shows fast convergence, but its best accuracy remains slightly lower at 0.9737.

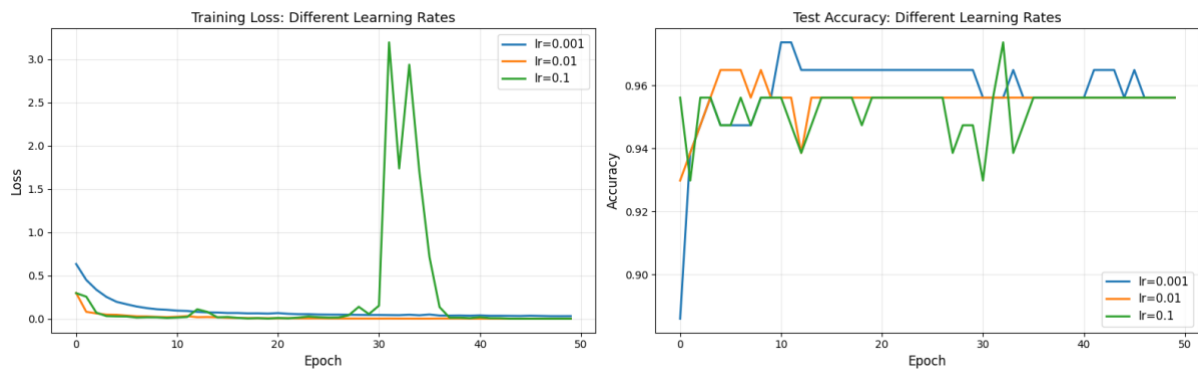
A brief summary is provided below.

- hidden_dim = 16: final accuracy 0.9561, best accuracy 0.9737
- hidden_dim = 32: final accuracy 0.9649, best accuracy 0.9825
- hidden_dim = 64: final accuracy 0.9649, best accuracy 0.9737

Overall, hidden_dim = 32 achieves the best performance and represents the most effective configuration among the tested settings.

Q2. Change the learning rate (lr) to 0.001, 0.01, and 0.1, and compare training stability.

The MLP was trained with learning rates 0.001, 0.01, and 0.1 to compare convergence behavior and stability. The training loss and test accuracy curves are shown below.

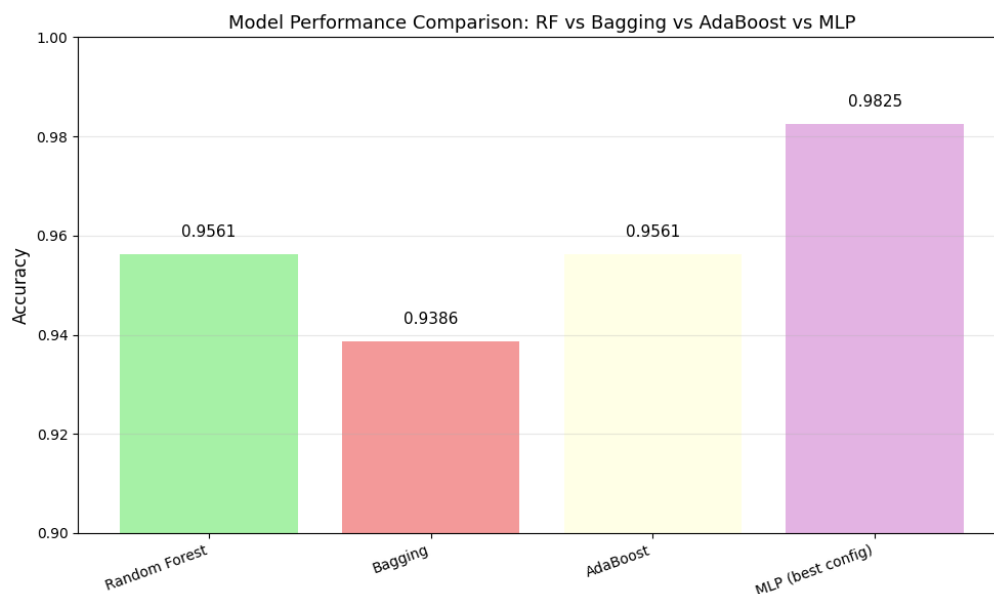


All three settings reach a similar final test accuracy of 0.9561, but their training dynamics differ. With $lr = 0.001$, the loss decreases gradually and smoothly, indicating stable but relatively slow convergence. With $lr = 0.01$, the loss drops much faster and remains low without noticeable oscillation, providing both quick convergence and stable accuracy. With $lr = 0.1$, the model initially converges quickly but shows large loss spikes around the middle of training before recovering, which suggests occasional overshooting of the optimum.

Overall, $lr = 0.01$ offers the best balance between convergence speed and stability, while $lr = 0.001$ is slower but safe, and $lr = 0.1$ is more unstable despite reaching the same final accuracy.

Q3. Discuss which model—among Decision Tree, Random Forest, Bagging, AdaBoost, and MLP—is the most suitable for this dataset.

The final accuracies of the four models are shown below, and a visual summary is provided in the following figure.



Random Forest and AdaBoost both achieve an accuracy of 0.9561, showing strong and stable performance on this dataset. Bagging performs slightly lower at 0.9386 but still provides reasonable results. The MLP, with its best configuration, achieves the highest accuracy of 0.9825, outperforming all ensemble-based models.

- Random Forest: 0.9561
- Bagging: 0.9386
- AdaBoost: 0.9561
- **MLP: 0.9825**

These results indicate that the MLP is the most suitable model for this dataset. Its higher capacity allows it to capture more complex nonlinear relationships compared to the tree-based ensemble methods. Random Forest and AdaBoost remain strong alternatives due to their robustness and ease of use, but the neural network ultimately provides the best overall performance.